MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU–UGP (HONOURS) REGULAR EXAMINATION MARCH 2026
FOURTH SEMESTER
MG4DSCBIF200-MOLECULAR STRUCTURES IN BIOINFORMATICS PERSPECTIVE
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest() and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

Short Answer Questions
Answer any 5 out of 7 Questions
Each question carries 2 marks

1. Commend on protein structure prediction.[Ap][CO1]

2. Acknowledge homology modeling? [S][CO2]

3. Acknowledge secondary structure of protein? [S][CO2]

4. Commend on Ramachandran plot? [Ap][CO1]

5. Acknowledge any two secondary structure prediction tools. [S][CO2]

6.Comprehend RNA secondary structure prediction? [A][C04]

7. Write any two protein structure validation tools.[A][C03]

[2 x 5 = 10]

Part B

Short Essay Questions
Answer any 4 out of 6 Questions
Each question carries 5 marks

8. Commend on the gap between protein sequence and structure. [Ap][CO1]

9. Commend Phi and Psi angles and their significance in protein structure. [Ap][CO1]

10. Acknowledge Chou–Fasman method for secondary structure prediction. [S][CO2]

11. Demonstrate homology modeling steps in brief. [S][CO2]

12. Write a note on RNA secondary structure prediction using MFold. [A][CO4]

13. Outline protein structure validation and its importance. [An][C03]

[5 x 4 = 20]

Part C

Long Essay Questions
Answer any 2 out of 4 Questions
Each question carries 10 marks

14. Assemble secondary structure prediction methods and tools. [S][C02]

15. Plot glycomics tools in detail. [A][C05]

16. Acknowledge tertiary structure prediction approaches including ab initio and threading methods. [S][C02]

17. Write a note on nucleic acid structure prediction and tools. [A][CO4]

[10 x 2 = 20]

END OF QUESTION PAPER